

Homework 3 Stat 603 Due Friday, February 7, 2025

1. Page 82, Problem 3.10

2. Page 82, Problem 3.15

3. Page 82, Problem 3.16

4. Page 83, Problem 3.18

Please see the attached page for the above four problems. Five points each for these four problems.

5. Refer to the student alcohol data in Homework #2, Problem 1.14. Let us focus on the students who took math (file: `student.math.csv`), and their final grade (G3) and absences. The final grade (G3) is on a scale 0-20. Convert that to a percentage score, and assume that grade A is 80% or above.

(a) Find the proportion (probability) of students who got an A in math.

(b) Find the proportion (probability) of students who had 10 or more absences.

(c) Find the probability of the intersection of the two events in (a) and (b).

(d) Find the probability of the union of the two events in (a) and (b).

(e) Find the conditional probability of a student getting an A grade if the student has 10 or more absences.

Do problem 5 in a Jupyter notebook. Ten points for this problem.

Computing Probabilities by Counting Outcomes

3.9 Assume each outcome in Ω has the same probability. In this case, show

$$P(\mathcal{E}) = \frac{\text{Number of outcomes in } \mathcal{E}}{\text{Total number of outcomes in } \Omega}$$

3.10 You roll a fair four sided die, and then a fair six sided die. You add the numbers on the two dice. What is the probability the result is even?

3.11 You roll a fair 20 sided die. What is the probability of getting an even number?

3.12 You roll a fair five sided die. What is the probability of getting an even number?

3.13 *I am indebted to Amin Sadeghi for this exercise.* You must sort four balls into two buckets. There are two white, one red and one green ball.

- (a) For each ball, you choose a bucket independently and at random, with probability $\frac{1}{2}$. Show that the probability each bucket has a colored ball in it is $\frac{1}{2}$.
- (b) You now choose to sort these balls in such a way that each bucket has two balls in it. You can do so by generating a permutation of the balls uniformly and at random, then placing the first two balls in the first bucket and the second two balls in the second bucket. Show that there are 16 permutations where there is one colored ball in each bucket.
- (c) Use the results of the previous step to show that, using the sorting procedure of that step, the probability of having a colored ball in each bucket is $\frac{2}{3}$.
- (d) Why do the two sorting procedures give such different outcomes?

The Probability of Events

3.14 You flip a fair coin three times. What is the probability of seeing HTH? (i.e. Heads, then Tails, then Heads)

3.15 You shuffle a standard deck of playing cards and draw a card.

- (a) What is the probability that this is a king?
- (b) What is the probability that this is a heart?
- (c) What is the probability that this is a red card (i.e. a heart or a diamond)?

3.16 A roulette wheel has 36 slots numbered 1–36. Of these slots, the odd numbers are red and the even numbers are black. There are two slots numbered zero, which are green. The croupier spins the wheel, and throws a ball onto the surface; the ball bounces around and ends up in a slot (which is chosen fairly and at random).

- (a) What is the probability the ball ends up in a green slot?
- (b) What is the probability the ball ends up in a red slot with an even number?
- (c) What is the probability the ball ends up in a red slot with a number divisible by 7?

3.17 You flip a fair coin three times. What is the probability of seeing two heads and one tail?

3.18 You remove the king of hearts from a standard deck of cards, then shuffle it and draw a card.

- (a) What is the probability this card is a king?
- (b) What is the probability this card is a heart?

3.19 You shuffle a standard deck of cards, then draw four cards.